

Werner Bisschoff

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Cape Town, Western Cape, South Africa

Systems Engineer

Professional Summary

Software engineer with 5+ years of experience bridging hardware and full-stack development. I combine technical depth in C++ and Python with a rigorous, agentic engineering methodology. My focus is on resilient system design and productivity-enhancing AI workflows, governed by security, processes, and specification-driven development.

Experience

Founder and Host

Cape Town

Divergent Tabletop

Jul 2025

- Founded and facilitate a neurodivergent-focused peer community, managing event operations and a WhatsApp-based community.
- Conflict resolution and management in a neurodiverse context.
- Applied systems thinking to solve community management challenges through software solutions.

Full-Stack Software Engineer

Cape Town

FARO Africa

Aug 2024 – Nov 2025

- Extended ERPNext using Python/JavaScript to improve workflows, pricing logic, and operational reporting (SQL) → reduced manual reporting time and improved data accuracy
- Built mobile application in Expo, including NFC (ISO 14443-4 APDUs) for e-paper price tags and card operations → enabled real-time price updates with fewer tagging errors
- Migrated internal mobile Retool workflows to Expo → improved performance and enhanced long-term maintainability
- Developed and maintained C# APIs supporting internal systems.
- Provisioned AWS infra with Pulumi and deployed services including Inngest and PayloadCMS.
- Diagnosed and resolved issues in a large existing ERPNext installation.
- Introduced LLM-assisted development workflows → improved debugging speed and code review throughput

Embedded Software Engineer

Remote Work

Ingenics Digital GmbH (through ViVa Outsourcing)

Mar 2023 – May 2024

- Designed an event-driven finite state machine for an I2C-based embedded system using C++ and FreeRTOS → created maintainable in-house architecture leading to fewer bugs and quicker development
- Developed ESP32 applications using C/C++ and ESP-IDF
- Integrated a configurable low-energy BLE stack for device communication
- Built Python-based tooling for serial/BLE communication, including client-facing test executables → accelerated testing and debugging workflows

- Created Python hardware mocks for rapid iteration and early-stage testing → enabled faster development cycles with fewer hardware dependencies
- Managed fast, reliable data interchange using a compact TinyFrame binary protocol
- Implemented asynchronous communication workflows with Python and Pytest
- Developed an active object within the QP Real-Time Embedded Framework (with QSPY) to simulate device behaviour
- Integrated a configurable BLE stack for device communication and implemented FOTA firmware updates over BLE

Software Developer

Century City, Cape Town

UMAN Technologies

Mar 2021 – Dec 2022

- Creating and maintaining Docker containers for development and CI/CD testing → improved development environment consistency
- Implementing and testing new services using RPC based on the SOME/IP protocol, as well as using perf to reduce performance bottlenecks
- Implementing IPC/RPC in existing C++ programs and Python scripts using Cap'n Proto and pycapnp
- Implementing a node tree to expose process-related variables and function calls to the IPC interface
- Analysing TCP/UDP traffic with Wireshark
- Leading a small team using AGILE development practices, including onboarding and mentoring new software developers

Junior Lecturer

Potchefstroom

North-West University

Feb 2020 – Dec 2020

- Lecturing Python and C++ programming for Introduction to Programming for first year IT students in both remote and in-person settings

Education

B.Eng. Computer and Electronic Engineering

Potchefstroom

North-West University

2020

- Focus on embedded systems, software engineering, and electronic design
- Developing an Android app with Kotlin to emulate an ISO 14443 protocol-based NFC payment system
- Developing microcontroller logic with C and the STM32 system as well as utilizing STM32CubeMX
- Implementing a PID controller with an Arduino to control a DC motor's voltage and speed
- Cleaning and analysing data from large spreadsheets with Python and Pandas, utilizing linear regression, correlation and machine learning

Skills

Systems Architecture: Event-driven FSM, RPC/IPC (Cap'n Proto, SOME/IP), State machines, Node trees

Testing & Quality: Unit testing (Pytest), Hardware mocks, Test executables, Code coverage analysis

Development Infrastructure: Docker, CI/CD, Cloudflare, Terraform, Linux, WSL2

Agentic Engineering & AI Integration: Agentic workflows, Claude Code, Spec-Driven TDD, Prompt Engineering

Workflow Automation: ERPNext, Python tooling, Retool → Expo migration

Languages: C/C++, Python, JavaScript/TypeScript, SQL, Elixir

Embedded Systems: ESP32/ESP-IDF, FreeRTOS, NimBLE, QP RTOS, TinyFrame protocol

Web Development: ReactJS, AstroJS, Django REST API, TailwindCSS, Phoenix

Primary: C++, FreeRTOS, ESP32/ESP-IDF, NFC APDU (ISO 14443-4)

Secondary: NimBLE, UART/SPI/I2C, TinyFrame, Hardware-in-the-Loop (HIL) Testing

Foundational: Assembly, Digital Logic, System Modeling

Other: ERPNext (Integration-focused), Expo Mobile

Primary: Python, ERPNext/Frappe Framework, Expo (React Native), Inngest

Secondary: AWS, Pulumi (IaC), PostgreSQL (RLS), Docker

Foundational: SQL, Object-Oriented Design, Spec-Driven Development

Other: C++, FreeRTOS (Low-level integration support)

AI Policy

I treat AI as an agentic partner within a strict engineering framework, moving beyond “vibe-coding” to structured, audit-ready workflows:

Spec-Driven Development (SDD): Prioritize context-heavy specifications to ensure alignment; AI-generated outputs are treated as hypotheses validated by human-led TDD.

Data Security: Enforce mandatory sandboxing for all agentic projects; strictly prohibit credential access and utilize tool-based retrieval for scoped, anonymized data only.

Engineering Rigour: All AI-assisted output undergoes rigorous human-in-the-loop review to ensure it meets enterprise quality and compliance benchmarks.

Privacy by Design: AI workflows are architected to avoid direct database access; sensitive data remains behind scoped, tool-mediated retrieval with no standing credentials exposed to agents.

Job Target

Seeking a mid to senior software engineering role (remote preferred, hybrid max. 2 days/week on-site in Cape Town) with emphasis on embedded systems, backend services, or AI-assisted development workflows. Prefers small, focused teams with clear goals and engineering culture. Open to permanent or contract (3+ months) arrangements.